# DEPARTMENT OF COMPUTER SCIENCE & SOFTWARE ENGINEERING

## SENG241 | Software Requirement Engineering

# Sehhaty Platform

Date: 04//2025

**Project Description:**
"Sehhaty" is a digital healthcare platform developed by the Saudi Ministry of Health to enhance public access to health services through smartphones and tablets. The app allows users to book outpatient appointments at public and private healthcare facilities, renew prescriptions, and view their personal electronic medical records. It also supports telemedicine services, provides access to laboratory results and medical reports, and offers accurate information on diseases, medications, and preventive care.
Additionally, the platform helps users stay up to date with the National Immunization Program (including flu and COVID-19 vaccinations). By digitizing routine healthcare services, "Sehhaty" reduces the need for in-person visits, improves accessibility, and ensures a more connected and informed healthcare experience for citizens and residents across the Kingdom.

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**Stakeholders:**

1. **Saudi Ministry of Health** – Oversees the app's development, ensuring alignment with national digital health strategies.
2. **Healthcare Providers (Hospitals and Clinics)** – Integrate their systems with "Sehhaty" to enable appointment scheduling and data sharing.
3. **Technology Companies and Developers** – Responsible for developing the app's infrastructure, ensuring high performance, usability, and security.
4. **Regulatory and Oversight Bodies** – Ensure the platform complies with cybersecurity and healthcare standards.
5. **Health Insurance Companies** – May connect their services to the platform to streamline claims and approvals.

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**Users:**

1. **Citizens and Residents of Saudi Arabia** – Use the app for everyday health services like appointments, immunizations, and test results.
2. **Elderly and Chronic Patients** – Depend on the app for medication renewals, lab follow-ups, and continuous care.
3. **Pregnant Women and Children** – Benefit from access to maternal and child healthcare tracking features.
4. **Healthcare Practitioners** – Doctors, nurses, and lab staff access and update medical records in real time.

5. **International Visitors** – Can use "Sehhaty" to access emergency care and services when eligible.

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**Goals and Objectives:**

1. **Improving health awareness** –Educating users about basic health information, such as proper nutrition and the importance of physical activity.
2. **Physical activity tracking** – Enabling users to record and track their physical activities, assisting them in achieving their fitness goals.
3. **Personal health management** – Providing tools to manage chronic health conditions, such as diabetes and high blood pressure, by monitoring symptoms and medications.
4. **Providing medical consultations** – Offering access to online medical consultations with healthcare professionals.
5. **Promoting a healthy lifestyle** – Motivating users to adopt new healthy habits, such as drinking sufficient water and obtaining quality sleep, through challenges and rewards.

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**Functional Requirements:**

1. **User Authentication** – Secure login via ID/residency number with multi-factor authentication.
2. **Appointment Management** – Book, cancel, and reschedule outpatient appointments; view in-app calendar with reminders.
3. **Telemedicine Services** – Real-time medical consultations through video or chat interfaces.
4. **Medical Records Access** – View prescriptions, reports, vaccination history, and sick leave documents.
5. **Digital Health Wallet** – Store digital medical certificates and test results for easy access and sharing.
6. **Emergency Assistance (Asafny)** – Location-based access to emergency services.
7. **Health Monitoring** – Track vital signs and daily activity with support for public health initiatives like "Know Your Numbers."

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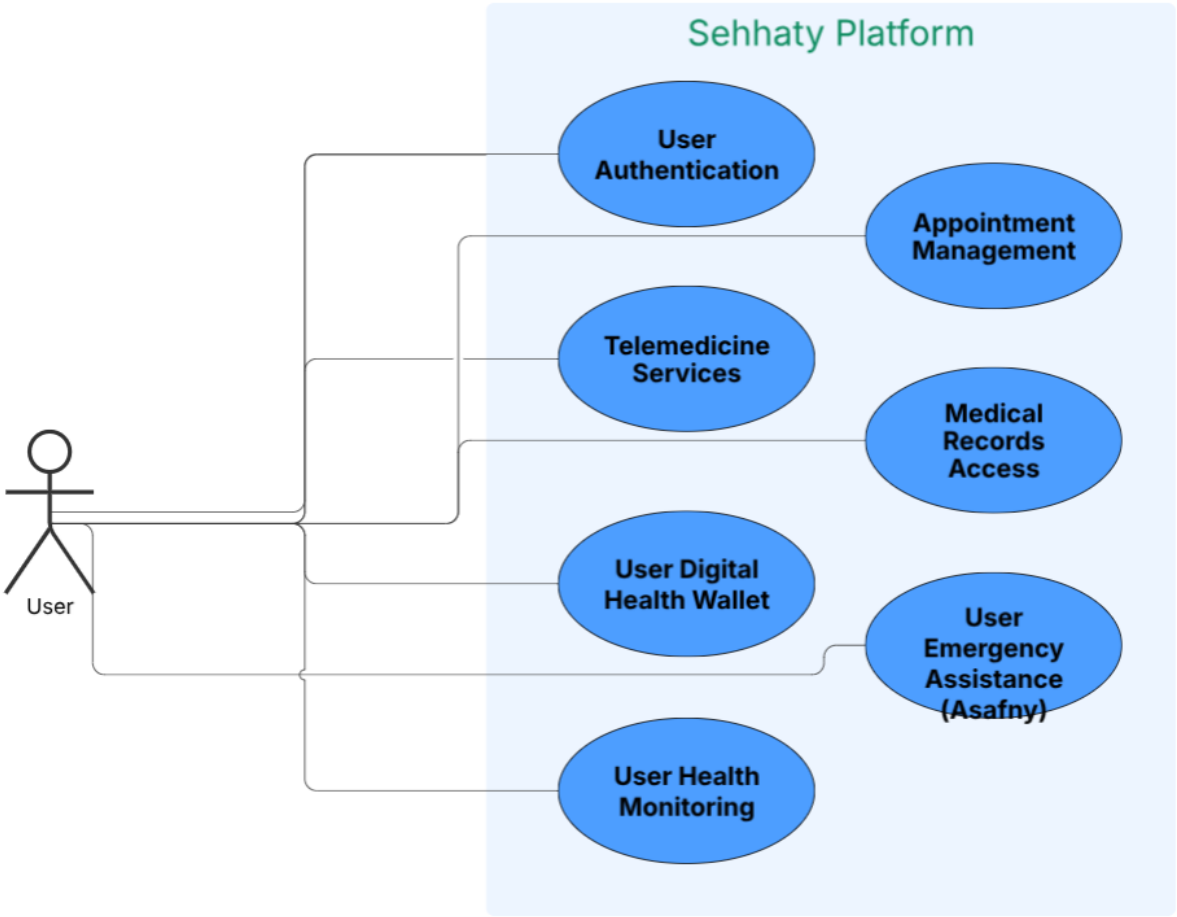
**Use Case Diagram:**

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**Non-Functional Requirements:**

1. **Performance** – App should load in under 3 seconds; response time under 2 seconds.
2. **Availability** – 99.9% uptime with recovery within 30 minutes of any failure.
3. **Security** – Strong encryption (e.g., AES-256); two-factor authentication and regular security updates.
4. **Compatibility** – Supports Android (10+) and iOS (14+); Arabic and English interface.
5. **Usability** – Intuitive design accessible to users with all levels of experience and ability.

6. **Scalability** – Should accommodate a 50% increase in user base without performance issues.
7. **Data Accuracy** – Health content must be reviewed quarterly by certified medical professionals.

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**Constraints:**

1. The total budget must not exceed $300,000.
2. The development timeline is limited to 5 months.
3. Must comply with Saudi healthcare data protection and cybersecurity regulations.

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**User Assumptions:**

1. Users own smartphones and are familiar with mobile apps.
2. Users are interested in tracking and managing their health digitally.
3. Users trust the platform's security and privacy.
4. Users prefer fast access to their health records and services.
5. Users are willing to share necessary data for medical purposes.

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